WEEK 3 · DAY 2 · INSTRUCTOR NOTES

Control Charts and Residuals

For the trainer only.

The one idea for today: *an alarm should fire when a reading is surprising, not when it is big.* A control chart learns "normal" from a quiet stretch; a residual compares each reading with what we expected.

At a glance

Length	~2 hours
Participants leave able to	build a control chart from a reference period, explain 3σ as a false-alarm budget, keep one set of limits per operating state, use rolling robust limits (median + MAD × 1.4826), compute and health-check a residual, and say why rolling detectors miss shifts and drift
Covers (original)	Week 4 Code Book: *1. Bus power*, *2. TWTA thermal*; Week 4 quiz topics **statistical process control** (Q7–12) and **residual-based detection** (Q13–18)
Datasets	`Time-series/week-04/data/bus_power_telemetry_hourly.csv`, `Time-series/week-04/data/twta_thermal_daily.csv` (the **week-04** version, which has the injected events)
Files you need open	animated slides, your **solution** notebook, the rule book PDF

🤫 **Don't show the label (answer-key) file today.** It's introduced on W4D1 when we score detectors. The checks know the answers without revealing the file.

Run of show (2 hours)

Time	Block	What happens	Slides
0:00	**Hook**	Mama Put's tomato prices: knowing "normal" lets you spot "strange"	1–2
0:08	**Control charts**	Anatomy → reference period → August dropout → why 3σ	3–7
0:25	**States & staleness**	Three operating states; the August band goes stale; rolling robust fix	8–11
0:40	**Tampering & run rules**	Common vs special cause; Rule 2 for level shifts	12–13
0:48	**Residuals**	Observed − expected; raw chart misses the spike, residual nails it; health checks; centred windows; three shapes	14–20

Time	Block	What happens	Slides
1:00	**Hands-on**	Part A (25 min), Part B (35 min). Walk the room.	21
1:45	**Wrap-up**	Scoreboards, recap, tease CUSUM	22

Explain it like this (plain English)

- **Control chart:** learn the average and the wobble (σ) from a calm stretch. Anything beyond 3 wobbles from the average is unusual.
- **Why 3σ:** it's a budget. On normal data, ±3σ is crossed by chance about 0.27% of the time (≈ 8 alarms in 2,880 hours). ±2σ: ≈ 131 alarms, and people stop listening.
- **Operating states:** eclipse, recharge and sunlit are three different "normals". Mix them and σ grows from 0.073 A to 0.594 A, and the chart goes blind. Fix: one set of limits per state. Not wider limits, not deleting eclipse hours, not 2σ.
- **Stale reference:** the array's normal output slowly climbs through the season (about 27.9 → 28.3 A daily average). A fixed August band flags 1,861 of 2,696 sunlit hours by November. Fix: rolling limits learned from the trailing week.
- **Robust:** the median and MAD ignore a few wild points; mean and σ don't. × 1.4826 converts MAD to the same size as σ.
- **Tampering:** reacting to everyday wobble as if it were a real change. Adjusting after every reading makes things worse (spread +39% in our simulation; theory says +41%).
- **Residual:** observed − expected. The amplifier runs hotter with higher drive and with the sun angle, so a big temperature isn't automatically a surprise. The recipe removes what we understand; what's left is the surprise.
- **The three shapes:** a residual catches **spikes**. A *fixed* recipe also shows a **level shift** (every day after 12 May is flagged), but a *rolling* "expected" swallows the shift in about three weeks and never sees a **slow drift**. That's tomorrow's CUSUM.

Numbers you'll quote (all computed from the data)

What	Value
Hours / states	2,880 hours: 46 eclipse, 138 recharge (3 h after each eclipse), 2,696 sunlit
Mean current by state	eclipse 0.03 A · sunlit 28.14 A · recharge 29.59 A
Reference period 1–15 Aug	360 hours, centre 27.87 A, σ 0.073 A, limits 27.65–28.09 A
σ over all sunlit hours incl. recharge	0.594 A (8× bigger)
Fixed August band, whole file	1,861 of 2,696 sunlit hours outside
Dropouts (in sunlit hours)	18 Aug 09:00 · 28 Sep 14:00 & 15:00 · 31 Oct 03:00, all ≈ 15.3–15.5 A (about 55% of normal)

What	Value
Rolling robust limits (168 h trailing, sunlit only)	5 flags = the 4 dropout hours + 1 false alarm (1 Sep 00:00)
The 1 false alarm	1 Sep 00:00 reads 29.48 A, a recharge-sized value with no eclipse flagged before it. Most likely the season's first, very short eclipse didn't show up in the hourly flag. A real "special cause", not a fault
Mean/σ rolling version	±2σ: 31 flags · ±3σ: 5 flags
σ right after the first 28 Sep dropout hour	plain 1.08 A vs robust 0.09 A
Tampering simulation	spread × 1.39
TWTA raw	mean 63.90 °C, σ 1.97 °C, 3σ ceiling 69.81 °C; spike 68.15 °C on 3 Sep 2024: **inside** the raw band
Recipe (refit without spike days)	temp ≈ 49.5 + 0.174 × drive + sun-angle wave; residual σ 0.22 °C
Spike as residual	+5.0 °C = 23σ on 3 Sep (15σ on 4 Sep)
Health checks	all 2024: lag-1 0.33, spread 0.22 / 0.49 → **not clean**; without the 2 spike days: lag-1 0.06, spread 0.22 / 0.22 → clean
After 12 May 2025	233 of 233 days beyond 3σ; mean residual +2.59 °C
Rolling 30-day median detector	spike 11 robust σ (caught); 0 flags after 12 May (shift swallowed in ~3 weeks)
Centred vs trailing 7-day window at the spike	centred mean residual +4.0 °C vs trailing median +5.3 °C

Slide talking points

#	Say this	Ask this
2	"Mama Put never studied statistics, but she has a control chart in her head."	"How do you know a price is too high?"
5	The original code book says σ ≈ 0.06 A. That's only true for a clean fortnight like this one.	–
6	18 Aug: half the normal output in full sun. NigComSat-1 was lost in 2008 to a solar-array problem; this is why we watch the array.	"Fault or normal?"
7	3σ is a budget, not a magic number.	"Would you answer 131 alarms in 4 months?"

#	Say this	Ask this
8	Click to reveal the rule: one set of limits per state.	"What happens to σ if we mix eclipse hours in?"
9	Not 1,861 faults. The *reference* is stale.	"Is the satellite broken in November?"
12	Tell the brake-pumping driver story.	"Who has worked with a boss who tampers?" 😄
16	The raw band has to be wide because temperature has known drivers.	"Can a 3σ line on raw temperature catch this spike?"
18	Click to show the table. Two bad days spoiled the recipe; the health checks caught it.	–
20	Cliffhanger: rolling detectors are blind to slow change.	"How would *you* catch a landlord who adds small-small?"

Answer key

Task	Answer	Common mistake → what to say
A1	`(bus["state"] == "recharge").sum()` → **138**	Counting eclipse hours (46) instead
A2	`ref.mean()` → **27.87**, `ref.std()` → **0.073**	Using the whole file instead of `ref`
A3	`centre - 3 * sigma` → **27.65**, `centre + 3 * sigma` → **28.09**	Giving ucl first (order is lcl, ucl)
A4	`outside.sum()` → **1,861**	Thinking it's wrong because it's huge. It's the lesson: stale reference
A5	`hours * 0.0027` → **7.8**, `hours * 0.0455` → **131**	–
B1	`(z.abs() > 3).sum()` → **5**	Forgetting `.abs()` (counts only one side)
B2	`"b"`, `"c"`	–
B3	`tw["twta_temp_c"] - expected` (on 3 Sep 2024: **+4.95 °C**)	Reversing the order (expected − observed)
B4	`"no"` (lag-1 0.33, spread 0.22 vs 0.49)	Saying "yes" because the mean is 0. One check isn't enough
B5	`residual[train].abs().idxmax()` → **2024-09-03**	Using `idxmin()`
B6	`residual2.loc["2024-09-03"] / sigma2` → **≈ 22.8**	Dividing by the raw σ (1.97)
B7	`(z2.loc["2025-05-12":].abs() > 3).sum()` → **233** (of 233)	–

Task	Answer	Common mistake → what to say
B8	`(rz.loc["2025-05-12":].abs() > 3).sum()` → **0**	Thinking 0 is a bug. It's the cliffhanger
🌶️	`tampered.std() / calm.std()` → **≈ 1.39**	–

Quiz link (original Week 4 quiz, `Time-series/MCQ/week-04-mcq.csv`)

Q	Tests	Where it's taught
7	Common vs special cause → *tampering*	slide 12, bonus
8	Control limits = 3σ around the reference mean	slide 4, A2–A3
9	3σ budget: ≈ 8 false alarms in 2,880 hours	slide 7, A5
10	Exclude eclipse hours from the reference	slide 8, A1
11	Run rule 2 → level shifts	slide 13
12	One chart (set of limits) per operating state	slide 8
13	Residual = observed − expected	slide 15, B3
14	Median + MAD: extreme points can't move them	slide 11, B2
15	MAD × 1.4826 ≈ σ	slide 11, B2
16	A centred window absorbs part of the spike (True)	slide 19
17	Rolling residual detector is useless on slow drift	slide 20, B8
18	"Values all above zero" is NOT a health check	slide 18, B4

Troubleshooting

Problem	Fix
A chart cell fails with the "Do A2 and A3 first" message	Normal: they skipped a task. Fill the blanks above
B-part `NameError: name 'bus' is not defined`	Runtime restarted. Run STEP 0 and Part A again
Someone's B6 is about 13	They used `residual` (spoiled recipe) instead of `residual2`
Fast finishers	Ask them to change the trailing window in B1 from 168 to 24 hours and see what happens to the false alarms

Notes on the original material

- The original says the sunlit baseline σ is ~0.06 A with a 27.6–28.0 A band. On the real file, all sunlit hours give σ 0.594 A (band 26.43–29.99 A) because recharge hours and the dropouts are mixed in. Only a clean reference fortnight gives a small σ (0.073 A). We teach exactly that difference.
- The original's Week 3 table says the TWTA file has no anomalies. That's the **week-03** copy; the **week-04** copy (used today) has the 3 Sep 2024 spike and the 12 May 2025 shift.
- The original names "four injected events" in the bus file in one place and three in another. There are **three events (four hours)**: 18 Aug, 28 Sep (two hours), 31 Oct.